Chapter 3: Network Protocols and Communications



Introduction to Networks

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Chapter 3

3.1 Rules of Communication

3.2 Network Protocols and Standards

3.3 Moving Data in the Network

3.4 Summary

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3.1 Rules of Communication

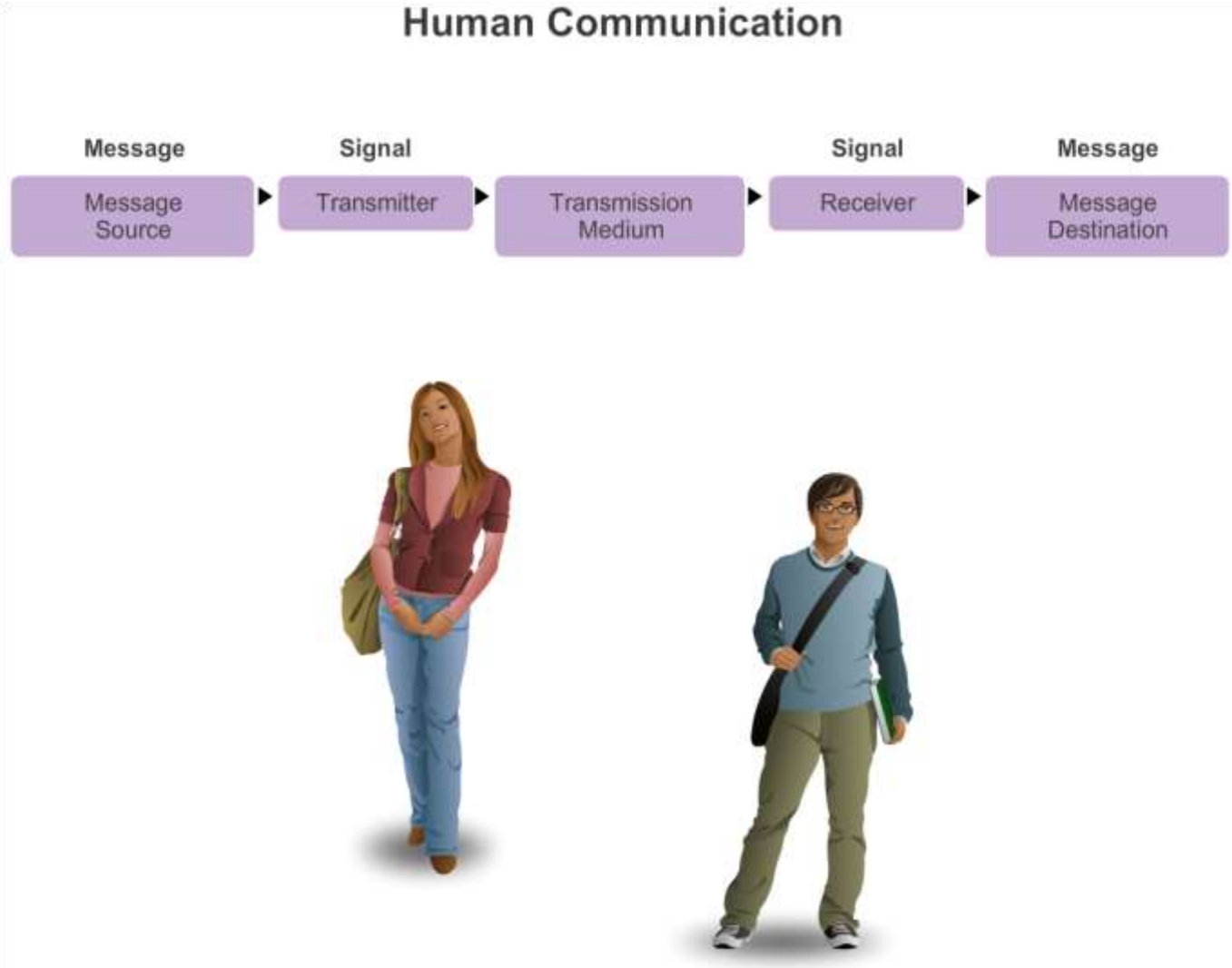


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The Rules

What is Communication?



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Establishing the Rules

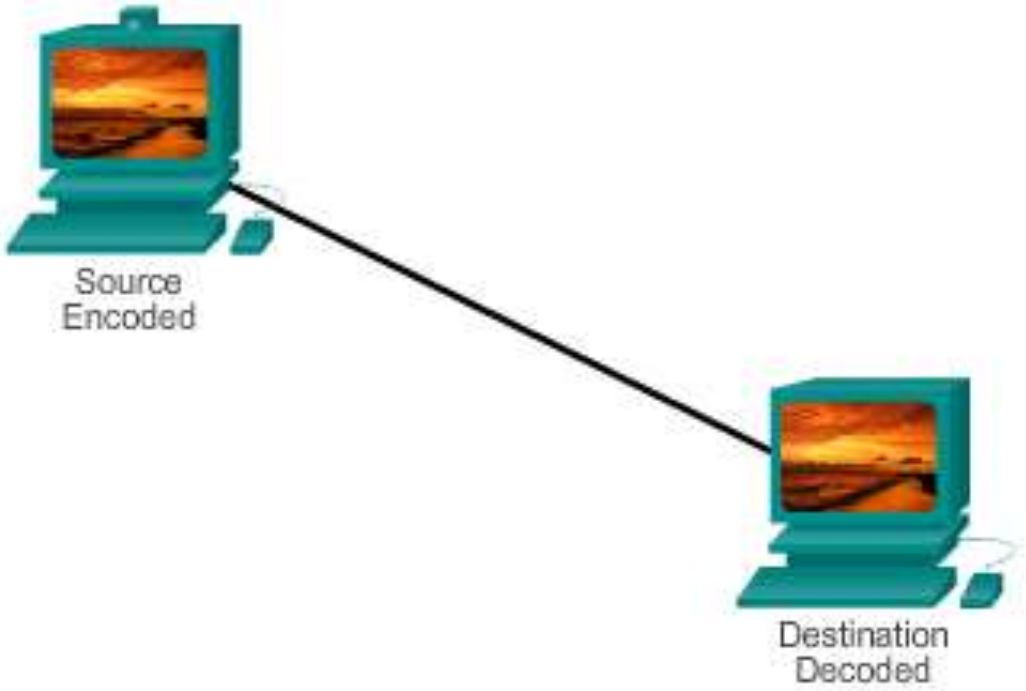
- An identified sender and receiver
- Agreed upon method of communicating (face-to-face, telephone, letter, photograph)
- Common language and grammar
- Speed and timing of delivery
- Confirmation or acknowledgment requirements

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The Rules

Message Encoding





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Message Formatting and Encapsulation

Example: Personal letter contains the following elements:

- Identifier of the recipient's location (network)
- Identifier of the sender's location (network)
- Salutation or greeting
- Recipient identifier (IP)
- The message content
- Source identifier (IP)
- End of message indicator

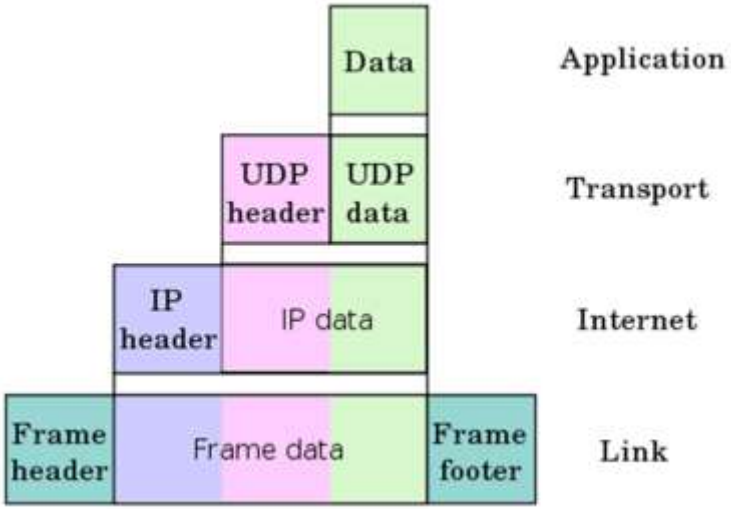


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Message Size

An overview of the segmenting process:

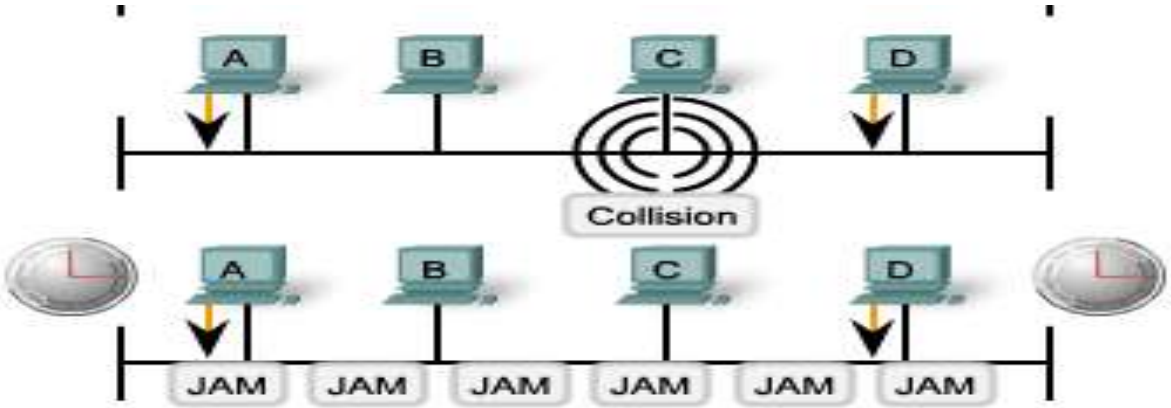
- The size restrictions of frames require the source host to break a long message into individual pieces (or segments) that meet both the minimum and maximum size requirements.
- Each segment is encapsulated in a separate frame with the address information, and is sent over the network.
- At the receiving host, the messages are de-encapsulated and put back together to be processed and interpreted.



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Message Timing

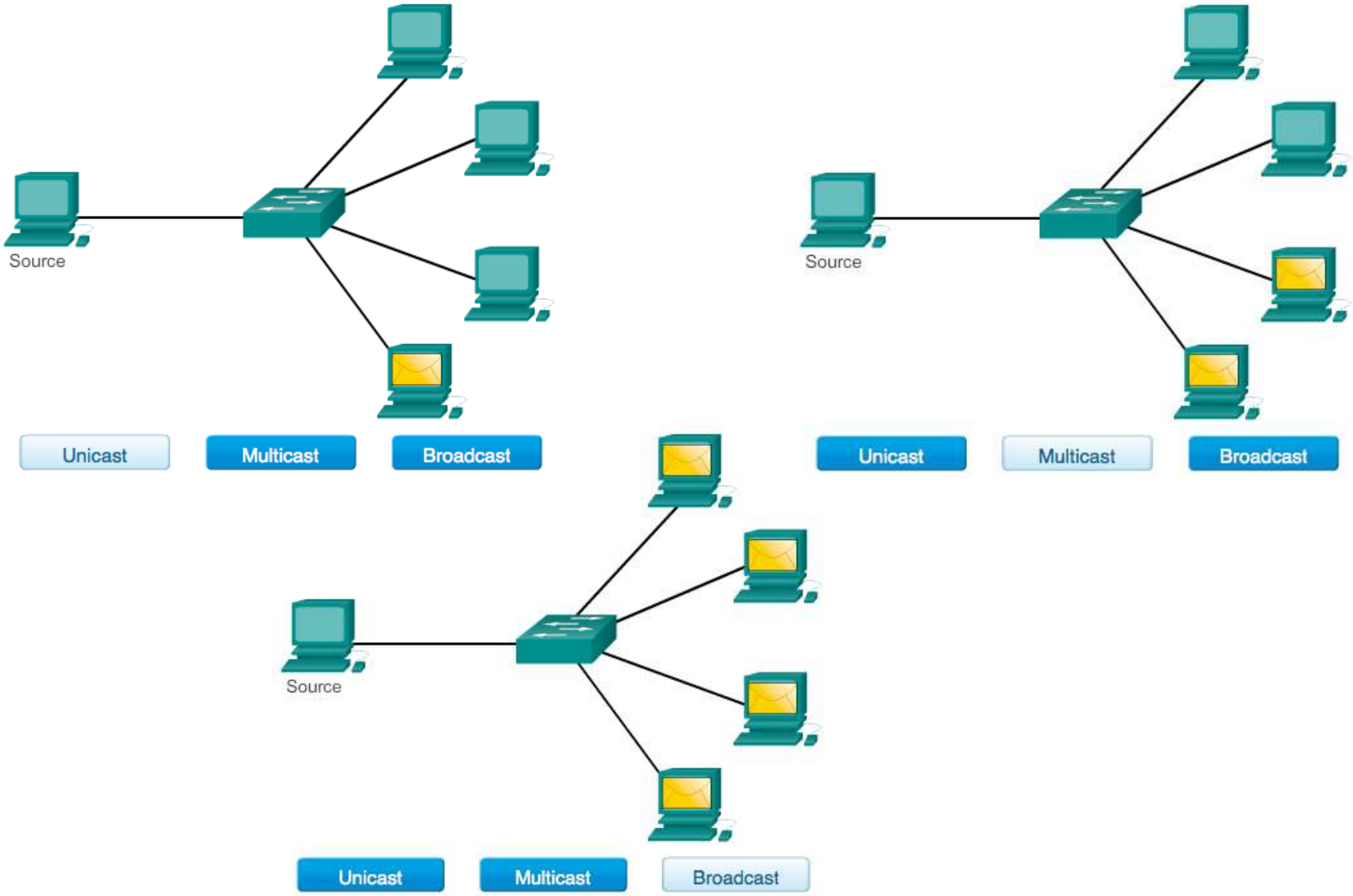
- Access Method (collisions can occur if everyone wants to talk at the same time)
- Flow Control (amount of data – windowing)
- Response Timeout (rules that specify how long to wait for responses and what action to take if a response timeout occurs)



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The Rules

Message Delivery Options



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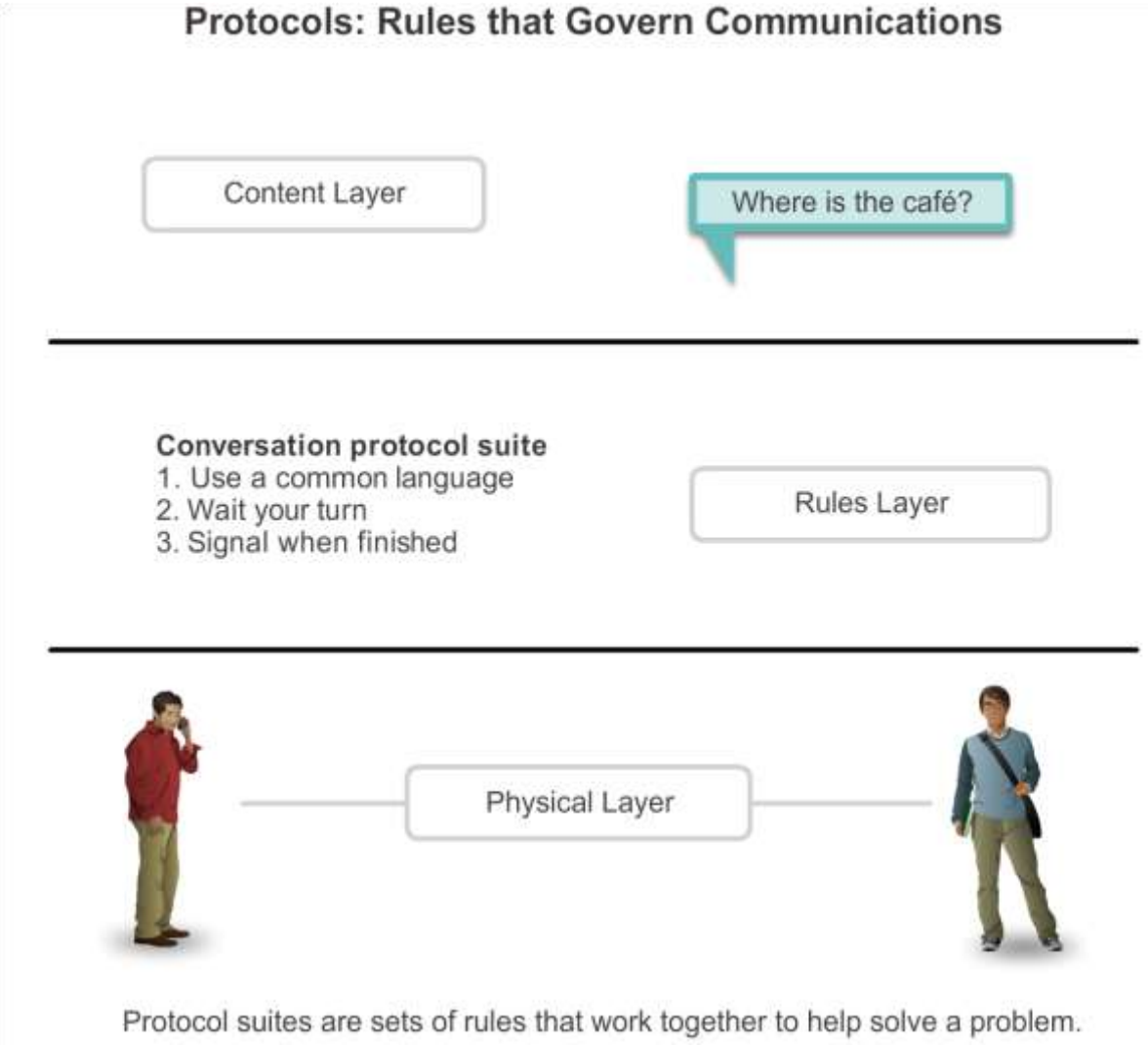
3.2 Network Protocols and Standards



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Rules that Govern Communications



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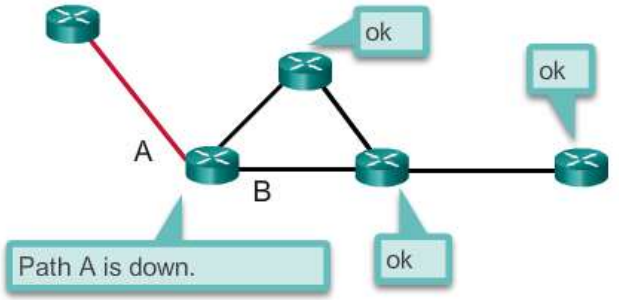
Network Protocols

A stack/suite of network protocols (examples of protocols are IP, HTTP, DHCP) describe:

1. How a message is formatted or structured



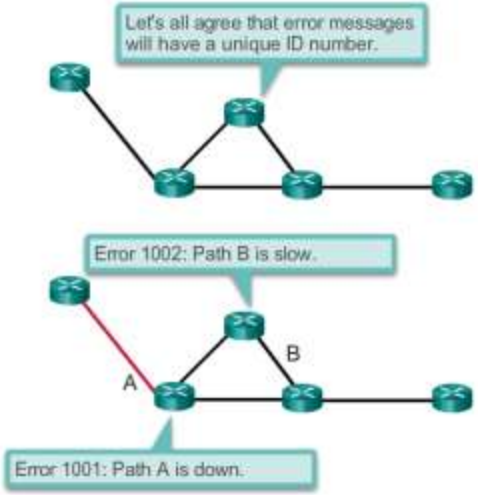
2. The process by which networking devices share information about pathways with other networks



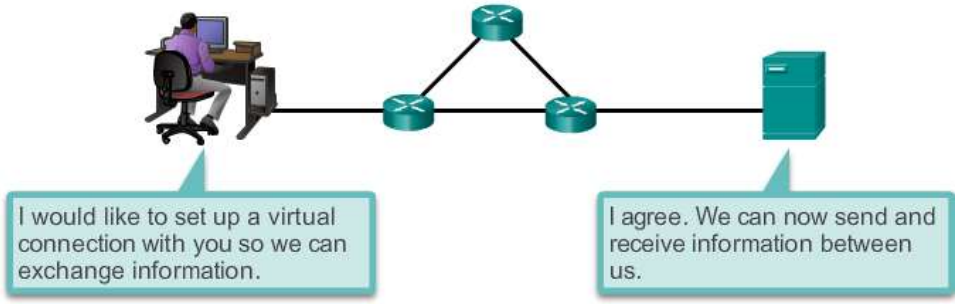
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Network Protocols

3. How and when error and system messages are passed between devices

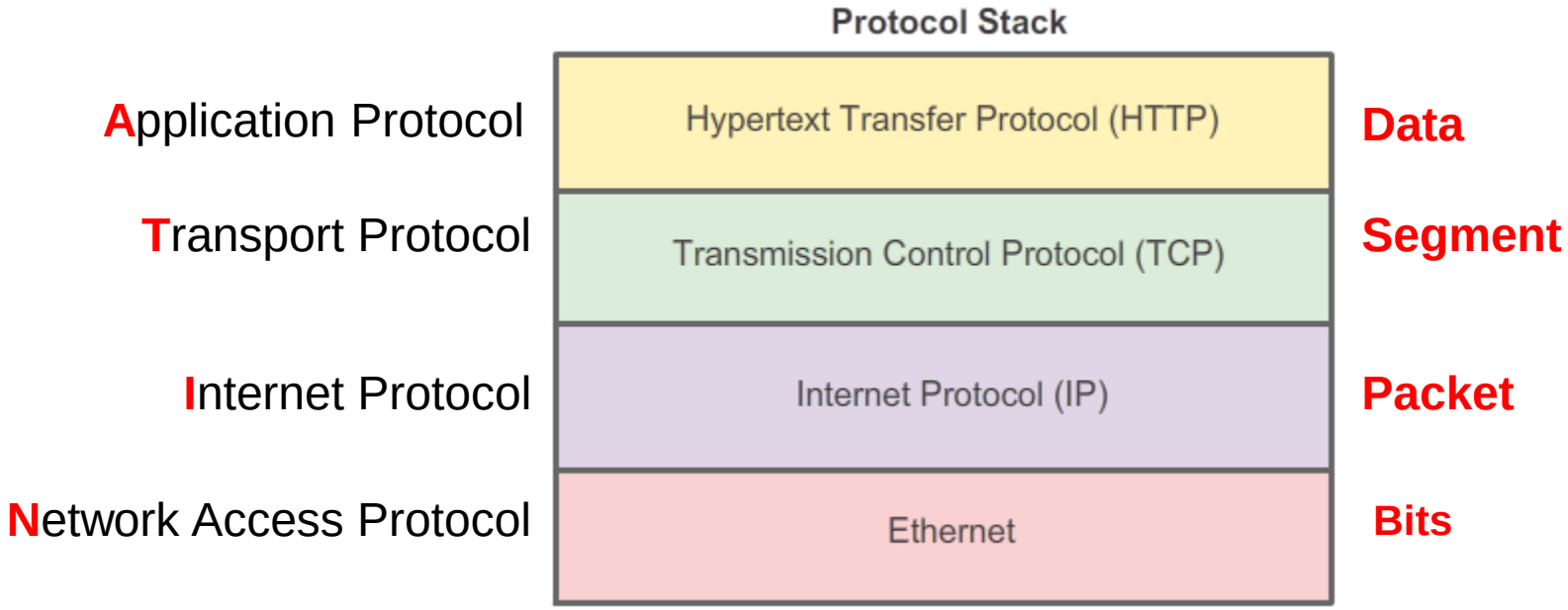


4.The setup and termination of data transfer sessions



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Interaction of Protocols



Application Protocol: governs the way a web client and web server interacts

Transport Protocol: divides HTTP segment into smaller pieces/segments

Internet Protocol: encapsulate segments into packets with IP addresses

Network Access Protocol: Format packets to be transmitted over media

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IEEE

The IEEE 802 family of standards deals with LANs and MANs

- Consist of 38 societies
- Published 130 journals
- Sponsor 1,300 conferences each year
- Has 1,300 standards and proje
- Has 400,000 members
- In 160 countries
- IEEE 802.3
- IEEE 802.11

IEEE 802 Working Groups and Study Groups

- 802.1 Higher Layer LAN Protocols Working Group
- 802.3 Ethernet Working Group
- 802.11 Wireless LAN Working Group
- 802.15 Wireless Personal Area Network (WPAN) Working Group
- 802.16 Broadband Wireless Access Working Group
- 802.18 Radio Regulatory TAG
- 802.19 Wireless Coexistence Working Group
- 802.21 Media Independent Handover Services Working Group
- 802.22 Wireless Regional Area Networks
- 802.24 Smart Grid TAG

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ISO



World's largest developer of international Standards as well as the OSI model

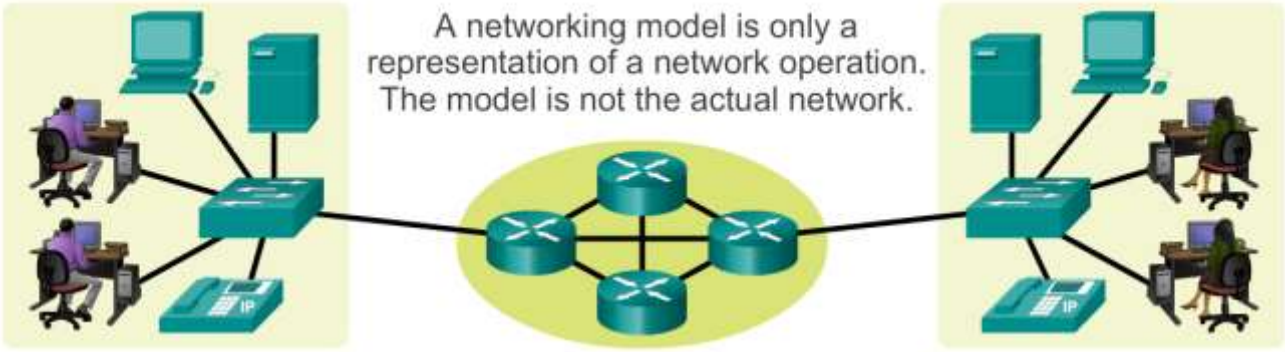
OSI Model



- ISO is best known for its OSI (Open System Interconnection) Model
- The OSI protocol suite was implemented on telecommunications equipment
- The TCP/IP protocol suite was implemented for use on the Internet

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Benefits of Using a Layered Model



OSI Model	TCP/IP Protocol Suite	TCP/IP Model
Application	HTTP, DNS, DHCP, FTP	Application
Presentation		
Session		
Transport	TCP, UDP	Transport
Network	IPv4, IPv6, ICMPv4, ICMPv6	Internet
Data Link	PPP, Frame Relay, Ethernet	Network Access
Physical		

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Other Standards Organization

- The Electronic Industries Alliance (EIA) - **electrical wiring and connectors**
- The Telecommunications Industry Association (TIA) – **develop communication in areas like cell, VoIP, satellite, radio**
- The International Telecommunications Union – Telecommunications Standardization Sector (ITU-T)- **divine standards for video compression broadband, DSL, internet protocol TV**
- The Internet Corporation for Assigned Names and Numbers (ICANN) - **coordinates IP address allocation, manage DNS, port numbers of TCP/UDP**

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The OSI Reference Model

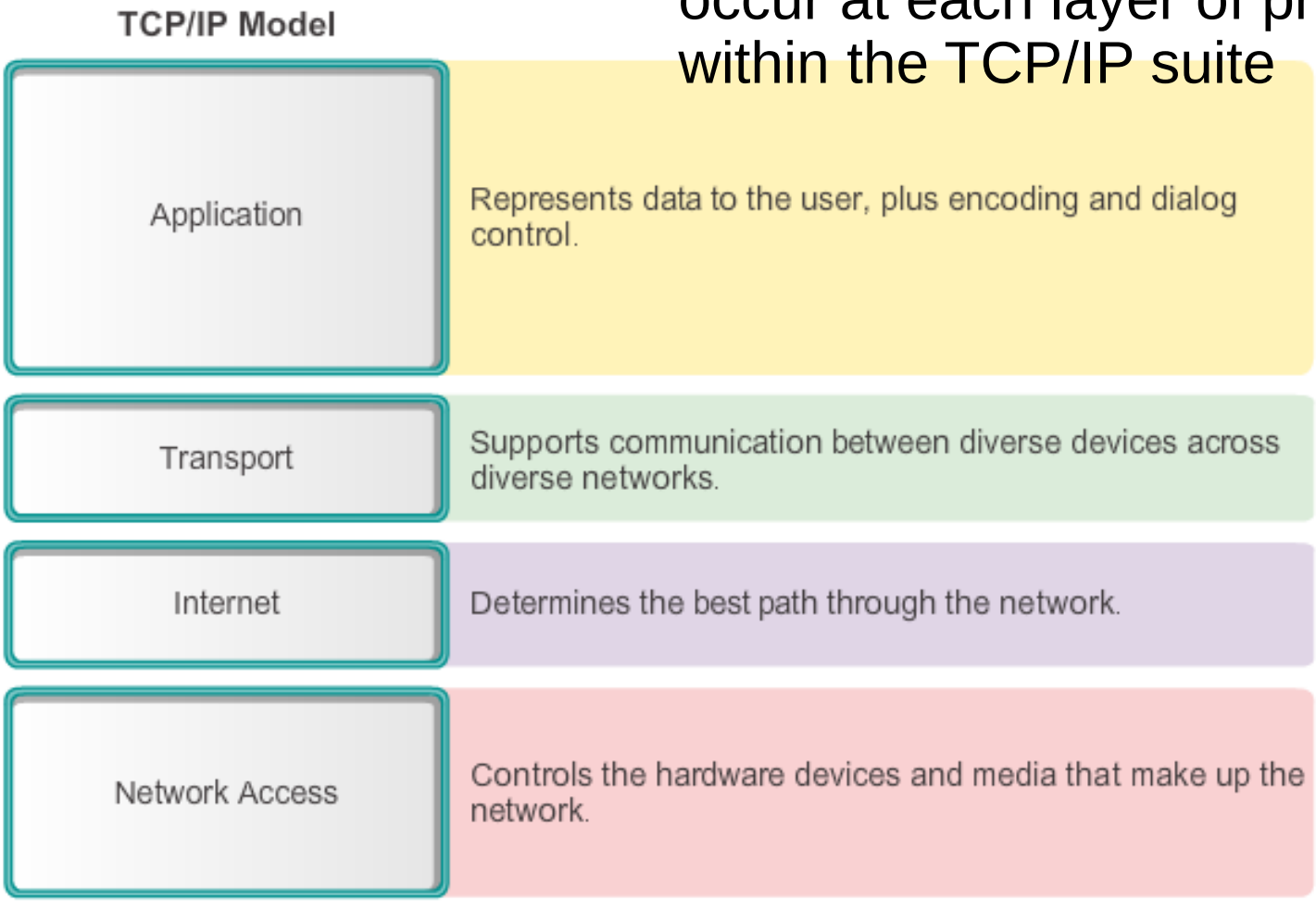
The primary purpose of a reference model is to get clearer understanding of the functions and processes involved.



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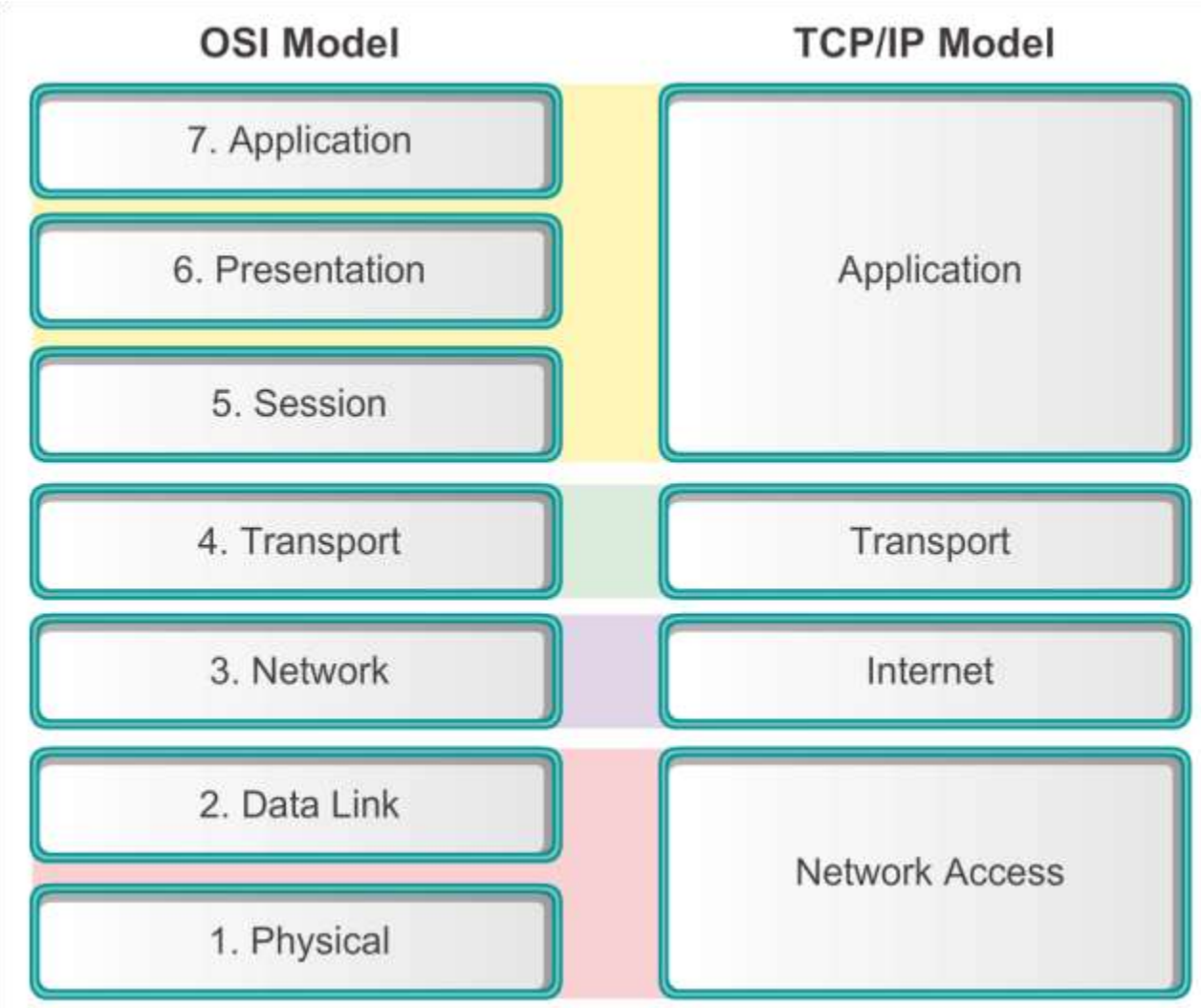
The TCP/IP Reference Model

describes the functions that occur at each layer of protocols within the TCP/IP suite



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Comparing the OSI and TCP/IP Models



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Comparing the OSI and TCP/IP Models

There are benefits to using a layered model to describe network protocols and operations. Using a layered model:

- **Assists in protocol design**, because protocols that operate at a specific layer have defined information that they act upon and a defined interface to the layers above and below.
- **Fosters competition** because products from different **vendors** can work together.
- Prevents technology or capability **changes** in one layer from **affecting other layers** above and below.
- Provides a **common language** to describe networking functions and capabilities.

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3.3 Moving Data in the Network



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Communicating the Messages

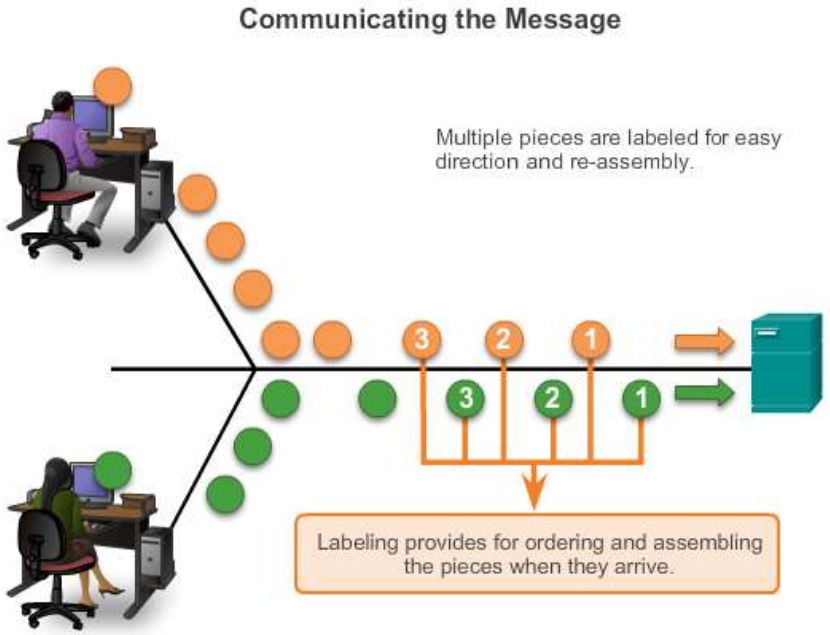
- Benefits of Message Segmentation

  Different conversations can be **interleaved/multiplexed**

  Increased reliability of network communications **(different parts/resend)**

- Segmenting message disadvantage

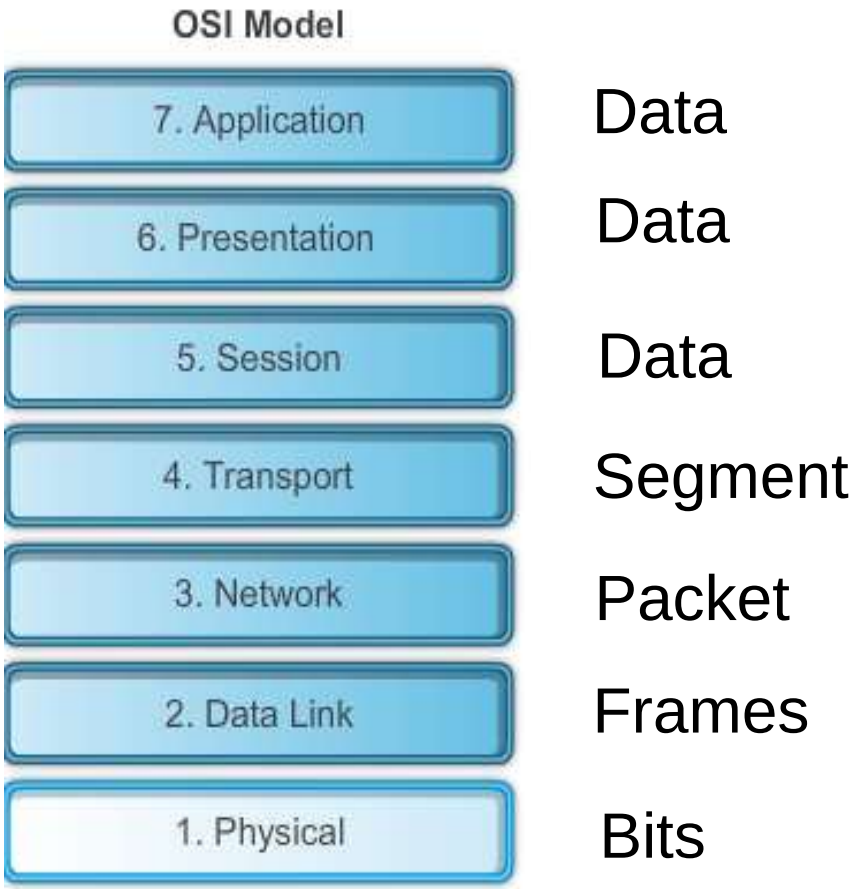
  Increased level of **complexity**



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Protocol Data Units (PDUs)

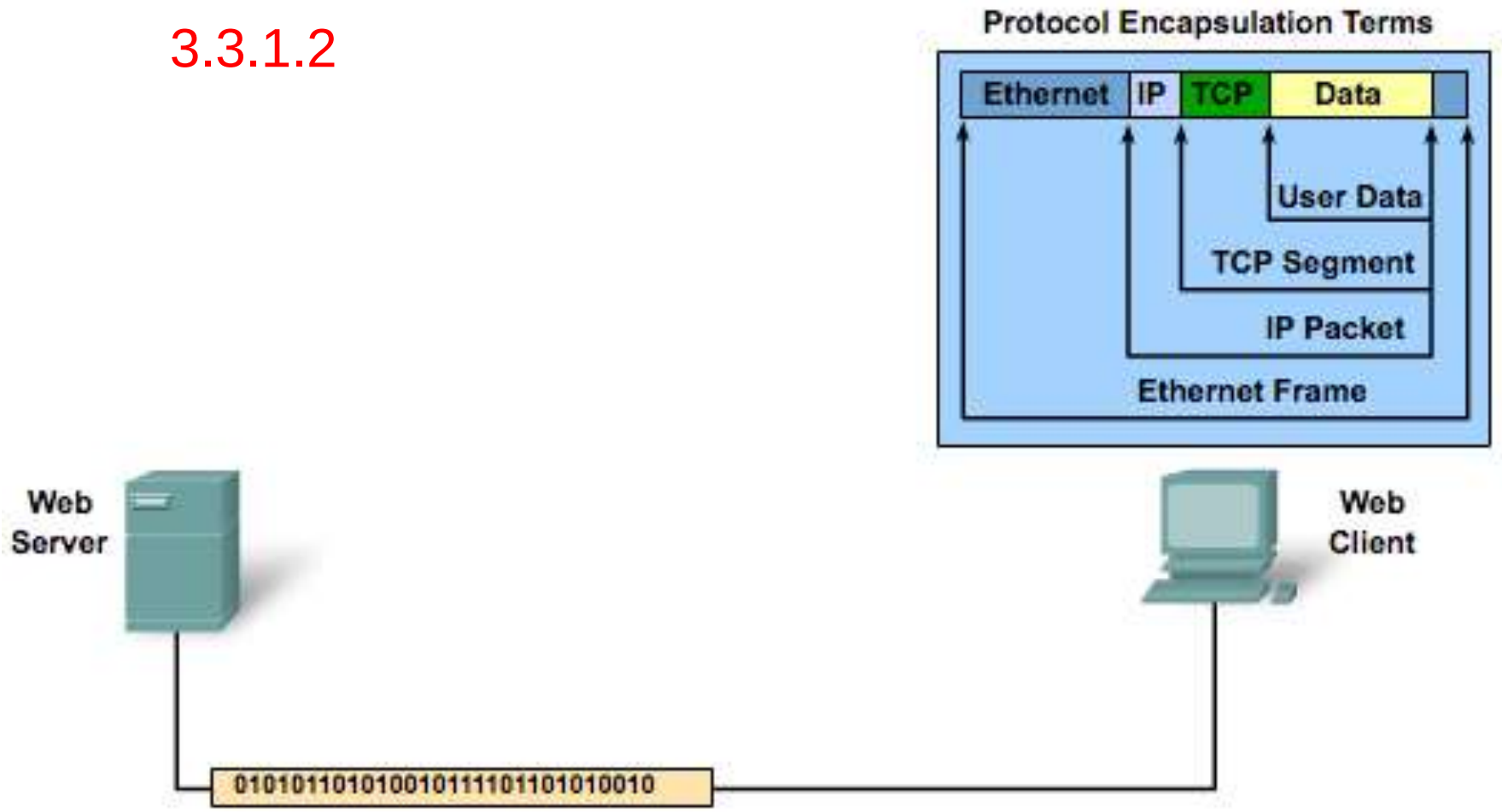
- PDUs at different levels are called:



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Protocol De-encapsulation

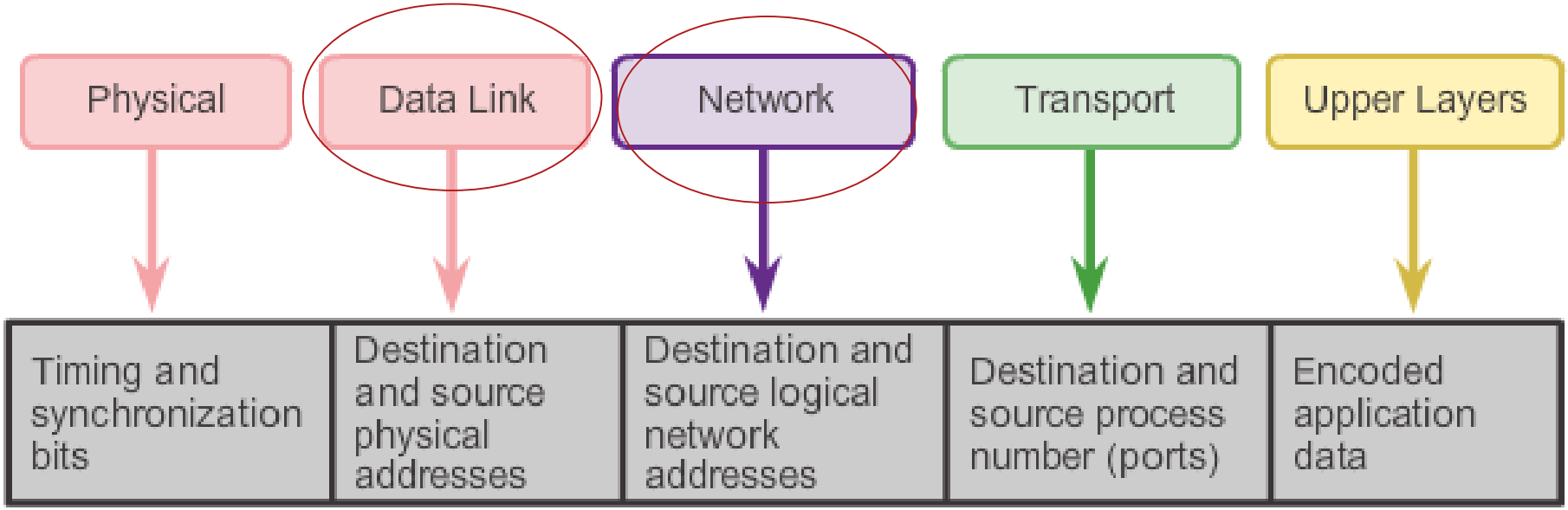
3.3.1.2



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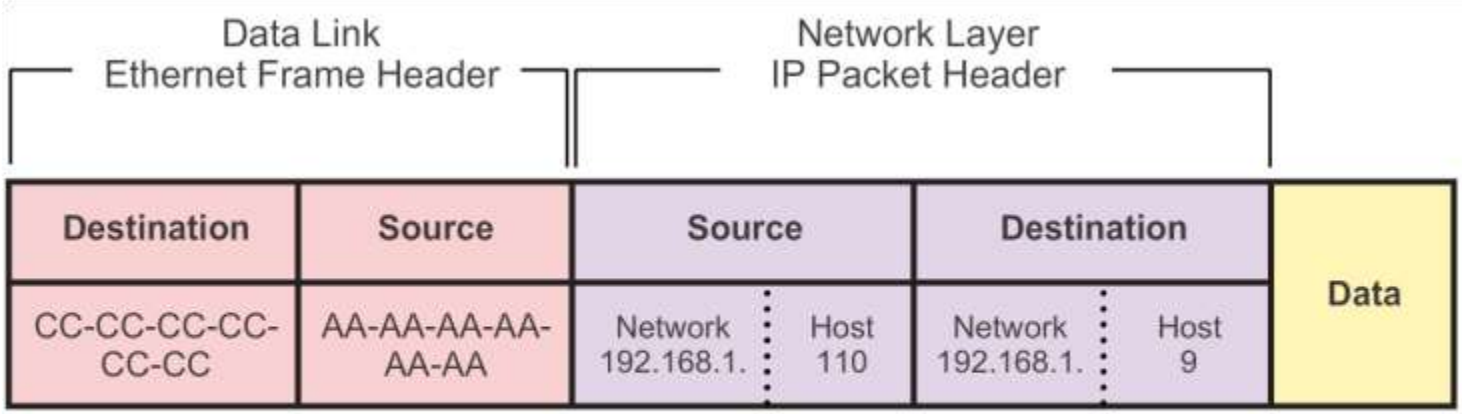
Accessing Local Resources

Network Addresses and Data Link Addresses

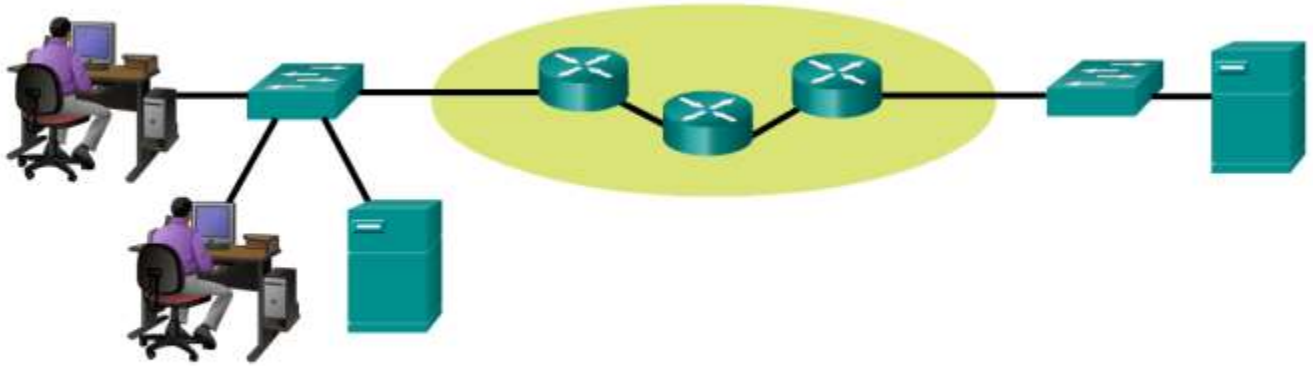


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Communicating with Device / Same Network



PC1
192.168.1.110
AA-AA-AA-AA-AA-AA



FTP Server
192.168.1.9
CC-CC-CC-CC-CC-CC

OSI Model

- 7. Application
- 6. Presentation
- 5. Session
- 4. Transport
- 3. Network
- 2. Data Link
- 1. Physical

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MAC and IP Addresses

- IPv4 Address: 32 bit address – 192.168.10.2

(11000000.10101000.00001010.00000010)

- IPv6 Address: 128 bit address

(0000000000000000.0000000000000000…) 8x

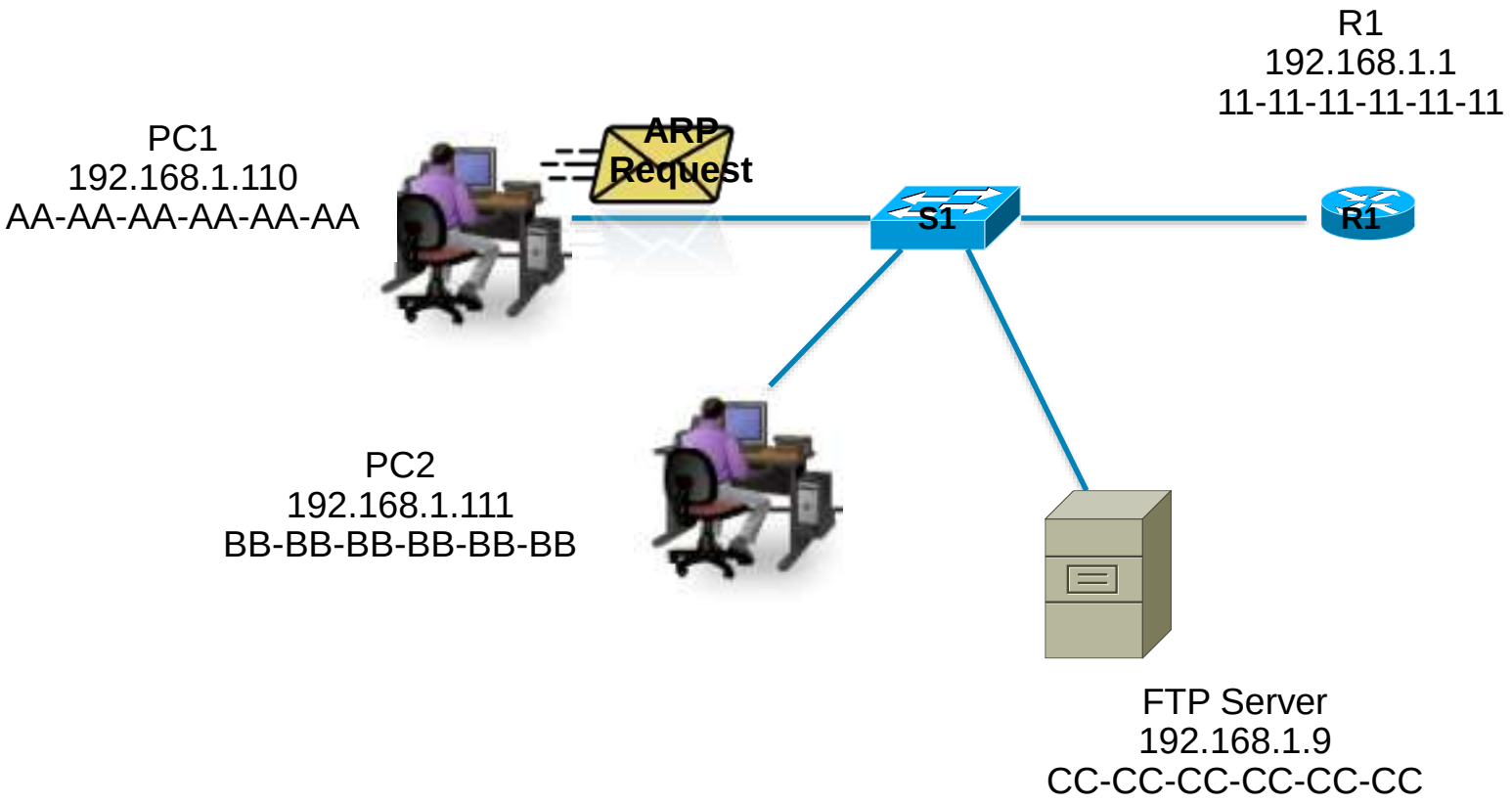
- MAC address/Burned-in Address (BID): 48-bit addresses

(1254.BC32.334B.CCB2.EEF2.FFD3)

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MAC and IP Addresses

PC1 broadcasts ARP request to LAN to find out to whom 192.168.1.111 belongs. PC2 will respond with ARP reply and its MAC address



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Communicating Device / Remote Network

Data Link
Ethernet Frame Header

Network Layer
IP Packet Header

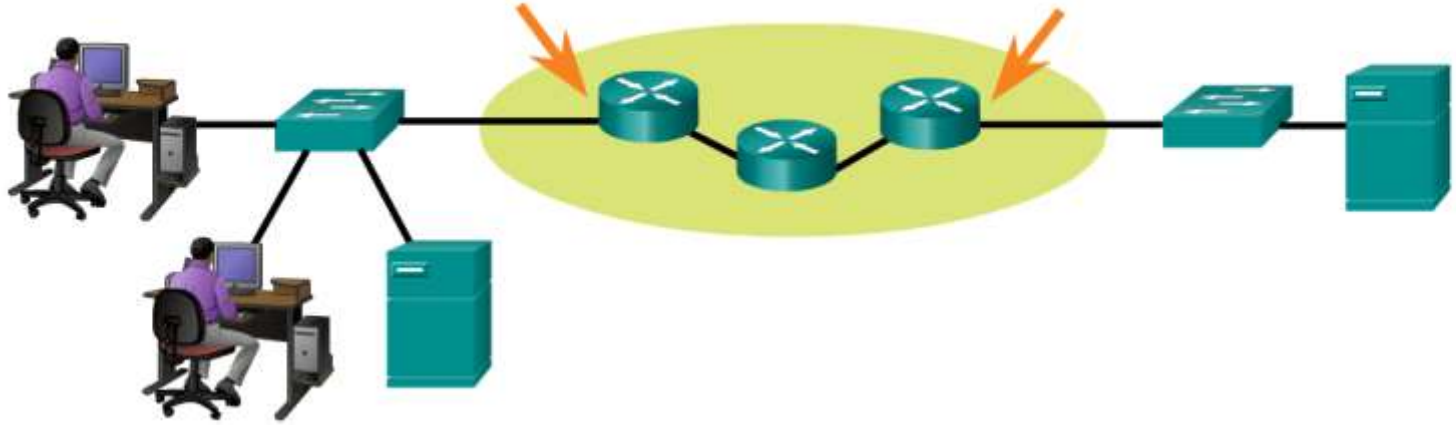
Destination	Source	Source		Destination		Data
11-11-11-11-11-11	AA-AA-AA-AA-AA-AA	Network 192.168.1.	Device 110	Network 172.16.1.	Device 99	

PC1
192.168.1.110
AA-AA-AA-AA-AA-AA

R1
192.168.1.1
11-11-11-11-11-11

R2
172.16.1.99
22-22-22-22-22-22

Web Server
172.16.1.99
AB-CD-EF-12-34-56



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Summary

In this chapter, you learned:

- Data networks are systems of end devices, intermediary devices, and the media connecting the devices. For communication to occur, these devices must know how to communicate.
- These devices must comply with communication rules and protocols. TCP/IP is an example of a protocol suite.
- Most protocols are created by a standards organization such as the IETF or IEEE.
- The most widely-used networking models are the OSI and TCP/IP models.
- Data that passes down the stack of the OSI model is segmented into pieces and encapsulated with addresses and other labels. The process is reversed as the pieces are de-encapsulated and passed up the destination protocol stack.

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Summary (cont.)

In this chapter, you learned:

- The OSI model describes the processes of encoding, formatting, segmenting, and encapsulating data for transmission over the network.
- The TCP/IP protocol suite is an open standard protocol that has been endorsed by the networking industry and ratified, or approved, by a standards organization.
- The Internet Protocol Suite is a suite of protocols required for transmitting and receiving information using the Internet.
- Protocol Data Units (PDUs) are named according to the protocols of the TCP/IP suite: data, segment, packet, frame, and bits.
- Applying models allows individuals, companies, and trade associations to analyze current networks and plan the networks of the future.

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Homework

- Activity 3.2.2.5
- Activity 3.2.4.5
- Lab 3.2.4.6 (building packets)
- Activity 3.3.1.5
- Activity 3.4.1.2 (wireshark)

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